1.2.1 Properties of Addition

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In class VI, we have studied that the collection of whole numbers is closed under the addition operation. The sum of two whole numbers is always a whole number. Does this property hold for the collection of integers also?

Complete the given table and check whether the sum of two integers is an integer or not?

(i) $7+(-5)=$	(ii) $(-6)+(-13)=$	(iii) $25+9=$
(iv) $(-12)+4=$	(v) $41+32=$	(vi) $(-19)+(-15)=$
(vii) $52+(-15)=$	(viii) $(-7)+0=$	(ix) $0+12=$
(x) $14+0=$	(xi) $(-6)+(-6)=$	(xii) $(-27)+0=$

We observe that in all the above cases the sum of two integers is an integer. Since addition of integers is an integer, we say that integers are closed under addition. This property is known as "closure property" of integers on addition.

Therefore, for any two integers a, b; $a+b$ is also an interger.

We observe one more property of integers. The order in which we add two integers does not matter. For example, $12+(-13)$ is the same as $(-13)+(12)$. Also $(-7)+(-5)$ is the same as $(-5)+(-7)$.

This property is known as "commutative property" of integers.

Therefore, for any two integers a, b; $a+b = b+a$.

What happens when we add three integers? For example, will $(-7)+(-2)+(-9)$ give the same value when they are added in any way of grouping.

Let us check by grouping the integers as [(–7)+(–2)]+ (–9) and (–7) + [(–2) + (–9)]. First let us find the value of [(–7)+(–2)]+ (–9).

[(–7)+(–2)]+ (–9) = (–9) + (–9) = –9 – 9 = –18

Let us illustrate this with the number line : $[(-7)+(-2)]+(-9)$ can be represented as

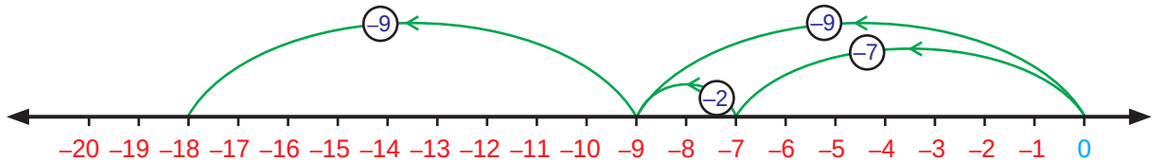


Fig. 1.9

[(–7)+(–2)]+ (–9) = –18

Then we will find the value of (–7) + [(–2) + (–9)]

(–7) + [(–2) + (–9)] = (–7) + (–11) = –7 – 11 = –18

(–7) + [(–2) + (–9)] can be represented on the number line as below.

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